Newton's laws of motion: applications 17a

NEWTON'S FIRST LAW: INERTIA Newton's first law of motion states that an object that is moving will continue to move at the same speed and in the same direction – unless a new force is acting on it. It also states that an object that is not moving will want to stay still – unless a new force is acting on it. This characteristic (not wanting to change) is called **inertia**.

1 In the following situations, state which object is changing its movement or direction (due to a force) and which is not (due to its inertia).

Situation	Changing	Not changing
A bus is taking off from the bus stop; a traveller falls backwards.		
A car is turning right; a passenger is pushed into the left door.		
The car stops suddenly; a bag is thrown against the windscreen.		
Bags fall off the luggage rack into the aisle when a train goes round a bend.		
A plane hits an air pocket; passengers float above their seats.		
A plane takes off; a passenger feels forced back into the seat.		
Your seat belt cuts into you as the car brakes suddenly.		
You sink to the floor as the lift goes up suddenly.		
The car slows down suddenly; the steering wheel pushes against the driver's chest.		
You fly over the handlebars because your bike hits a rock.		

2 Use Newton's law about inertia and your common sense to explain the following.

a People travelling in cars wear seatbelts (it's also the law!).

_____

b Motorbike riders don't wear seatbelts.

_____

c Commuters on city buses don't wear seatbelts, but people on overland coaches do.

_____

d Travellers on planes are asked to put their seatbelts on when the plane hits air turbulence.

_____

e The lift starts to go down quickly and you feel somewhat weightless.

_____

NEWTON'S SECOND LAW: F = m . a Newton's second law is about **acceleration**: acceleration makes an object go faster or slower, or change its direction. Acceleration depends on the force that is acting on the object and the mass of the object. Newton came up with the formula: Acceleration = force/mass or a = F / m.

3 State the units of the three quantities in the formula: a = acceleration: _____;

F = force: _____; m = mass: _____

4 A car's motor produces a force of 7.5 kN. The car has a mass of 1 500 kg. Calculate its acceleration. _____

5 A cyclist, having a mass of 100 kg (including the bike), is able to push on the pedals with a force of 200 N. Find how big the acceleration is. _____

6 Rearrange Newton's equation to make force the subject. F = _____

7 **a** Assume that your mass is 50 kg. On earth, acceleration due to gravity is 10 m/s^2. Calculate your weight force, i.e. your weight. (Scientifically, weight is measured in newtons, not kilograms.)

_____

© Chris Greef, Suada Bilali and Blake Education 2006

Newton's laws of motion: applications 17b

b At an altitude of 1000 km above the earth's surface, gravity acceleration is only 7.3 m/s^2. Calculate your weight there. _____

c At an altitude of 10 000 km above the earth's surface, gravity acceleration is only 1.5 m/s^2. Find your weight there.

_____

d On the moon, gravity acceleration is 1.6 m/s^2. Find your weight on the moon.

_____

e On the sun, gravity acceleration is about 270 m/s^2. Determine your weight on the sun.

_____

NEWTON'S THIRD LAW: ACTION – REACTION Newton's third law is about pairs of forces acting on objects. An action force is acting on one object in one direction and an equal but opposite reaction force is acting on another object in the opposite direction. In many cases, one of the forces is not obvious and may be overlooked entirely.

8 **a** You are holding a book on your hand in front of you. The weight of the book is the action force. In the box at right, sketch your hand with the book on it.

b Draw an arrow from the centre of the book downwards. Label it the action force.

c State what is providing the balancing force, ensuring that the book is not moving.

_____

d State how big this balancing force is compared to the action force. _____

e Draw another arrow to represent this balancing force on the book. Label it the reaction force.

f Imagine a friend adds ten books to the first one. State how the action force has changed.

_____

g Explain what would have to happen to the reaction force to keep the books stationary.

_____

Another force pair involves the gravity force of the books on the earth and the gravity force of the earth acting on the books. These forces act on different objects.

h Which gravity force, the books' gravity force or the earth's gravity force, is bigger? _____

j State what would happen if the earth's gravity force were bigger than your arm muscle force.

_____

9 **a** The drive wheel of a car is pushing against the bitumen of the road and the car moves forwards. In the box at right, sketch a car wheel on a road surface.

b Draw an arrow starting at the wheel and pushing backwards, representing the action force. Label it.

c Draw another arrow as the ground is pushing against the wheel, representing the reaction force. Label it.

d State how those two forces are similar and different.

_____

_____

© Chris Greef, Suada Bilali and Blake Education 2006